

PARIMAL GAVALI

Machine Learning & Applied AI | Backend Engineering

Bielefeld, Germany

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PROFESSIONAL PROFILE

Master's student in Informatics for Natural Sciences with nearly three years of professional backend-development experience in the banking domain. Experienced in building Java and Spring Boot microservices as well as applied ML and AI systems using Python, PyTorch and FastAPI. Areas of interest include production-oriented ML systems, document intelligence, NLP and time-series analysis.

PROFESSIONAL EXPERIENCE

Bielefeld University, Bielefeld

06/2026 - Present

Student Assistant - Web Development | Part-time, remote

- Contribute to the development and maintenance of BIIGLE, a web-based scientific application.
- Develop and improve user-interface components using Vue.js and related web technologies.
- Support REST API and backend functionality using Laravel.
- Investigate software defects, implement fixes and participate in Git-based code reviews.

Accenture Solutions, Pune, India

03/2023 - 09/2024

Application Development Analyst | Client: Commerzbank, Banking Domain

- Developed and maintained scalable Java and Spring Boot microservices supporting core banking workflows and internal applications.
- Translated complex business requirements into robust backend solutions in collaboration with business analysts, product owners and development teams.
- Implemented unit and integration tests using JUnit and Mockito, maintaining test coverage above 90%.
- Performed root-cause analysis for production incidents and implemented permanent fixes.
- Conducted code reviews, resolved SonarQube findings and applied clean-code and SOLID principles.

Accenture Solutions, Pune, India

12/2021 - 02/2023

Application Development Associate | Client: Commerzbank, Banking Domain

- Developed backend components for enterprise banking applications using Java and Spring Boot.
- Created and maintained RESTful services and supported their integration with existing systems.
- Created automated tests, investigated application logs and resolved identified defects.

PROJECTS

Clinical Decision Support System for Early Sepsis Prediction

04/2026 - 08/2026

Bielefeld University | Python, PyTorch, scikit-learn, MIMIC-IV

- Developed an end-to-end pipeline for early sepsis detection using longitudinal ICU patient time-series data.
- Trained and compared Random Forest, GRU and LSTM models for predictions up to 25 hours before sepsis onset.
- Developed temporal features using fixed and expanding observation windows.
- Evaluated performance under severe class imbalance using AUROC, AUPRC, sensitivity and specificity.

AI-Powered Document and Thesis Summarization Platform

08/2025 - 01/2026

University of Tartu | University project in collaboration with SEB Bank

- Developed a web application for processing and summarizing theses, reports and research documents.
- Implemented PDF, DOCX and TXT ingestion with automated generation of two- to five-page LLM summaries.
- Implemented interactive editing, alternative summaries based on user feedback and text-to-audio output.
- Containerized the application with Docker and configured Bitbucket Pipelines, end-to-end tests and load testing with Apache JMeter.

EDUCATION

Bielefeld University, Bielefeld

10/2024 - Expected 04/2027

M.Sc. Informatics for Natural Sciences

- Coursework: Machine Learning, Deep Learning, Data Mining, Big Data Analytics, Statistical Natural Language Processing, Algorithmic Stochastics and Semantic Web.

University of Tartu, Tartu, Estonia

08/2025 - 01/2026

Erasmus+ Exchange Semester, Computer Science

- Coursework: Statistical Machine Learning, Databases, Public-Key Cryptography and Applied Cryptography.

Savitribai Phule Pune University, Pune, India

08/2017 - 08/2021

Bachelor of Engineering, Electronics and Telecommunications

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

Machine Learning & Data: PyTorch, scikit-learn, TensorFlow, NumPy, Pandas, Matplotlib, time-series analysis

Applied AI & LLM: OpenAI API, LangChain, LangGraph, RAG pipelines, vector databases

Backend: FastAPI, Spring Boot, REST APIs, Microservices

Tools & Platforms: Docker, Git/GitLab, Bitbucket Pipelines, Postman, JMeter

Databases: PostgreSQL, Oracle

LANGUAGES

English: C1

German: A2